

The MOND Acceleration Scale as the Cosmic Dynamical Acceleration

A scaling-MOND cosmology, its over-constrained web, and three honest bridges toward a theory of the dark sector

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June 2026 · analysis assisted by Claude (Anthropic); all results reproducible from the cited scripts

Abstract. We examine the proposal that the MOND acceleration scale is the cosmic dynamical acceleration, $a_0 = (c/2)\sqrt{(G\rho_c)} = cH(z)/Z$ with $Z = 2\sqrt{(8\pi/3)} \approx 5.789$. Algebraically this is Milgrom's coincidence $a_0 \approx cH_0$ re-expressed through the Friedmann relation; its distinctive, coefficient-free consequence is that a_0 evolves as $a_0(z) = a_0(0) \cdot E(z)$. Fitting this to the 2026 data (SPARC, Vărașteanu, MUSE-DARK) gives an exponent $p = 0.80 \pm 0.17$ in $a_0 \propto E(z)^p$, excluding constant a_0 at 5σ and the matter-only $(1+z)^{3/2}$ branch at 5σ . The premise generates an over-constrained web (one number, five independent measurements that cohere), and bridges toward a relativistic dark-sector theory: we show, from the AeST action, that a_0 is absent from the linear cosmological perturbations, so a running a_0 leaves the CMB exactly invariant. We are explicit about the boundaries: the $O(1)$ coefficient is a posit, not a derivation; and the framework says nothing forced about the Standard-Model constants. The decisive test is one clean a_0 at $z > 2$.

1. The premise and its one novel framing

The empirical anchor is the MOND acceleration scale measured from 175 SPARC rotation curves, $a_0 = (1.1\text{--}1.2) \times 10^{-10} \text{ m/s}^2$ [McGaugh et al. 2016]. The proposal writes it as half the surface gravity of the cosmic free-fall density:

$$a_0 = (c/2) \cdot \sqrt{(G\rho_c)} = c^2/(2R), \quad R = \sqrt{(8\pi/3)} \cdot c/H = c \cdot t_{\text{ff}}$$

Via the Friedmann relation $H^2 = (8\pi/3)G\rho_c$, this equals $a_0 = cH/Z$ with $Z = 2\sqrt{(8\pi/3)}$. The $\sqrt{(8\pi/3)}$ factor is genuine Friedmann physics; the remaining factor of 2 — the Schwarzschild surface-gravity reading $c^2/2R$ — is the only un-derived element. The density / surface-gravity *form* appears to be novel (it is not the cH_0 or $\sqrt{\Lambda}$ form used in Milgrom's papers), but it is a re-expression of a known coincidence, not new physics, and the coefficient is not unique: the de-Sitter-horizon readings of Milgrom ($Z = 2\pi$) and Verlinde ($Z \approx 6$) fit the same data equally well.

2. The falsifiable prediction, confronted with data

Because a_0 tracks a cosmic density, it evolves. The prediction is coefficient-free — Z cancels in the ratio:

$$a_0(z) / a_0(0) = E(z) = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}.$$

Fitting $a_0(z) = A \cdot E(z)^p$ to the real compilation, marginalizing the normalization A , yields the result below. Constant a_0 ($p = 0$) and the matter-only branch (effectively $p = 3/2$) are each excluded at 5σ ; the premise ($p = 1$) is consistent at 1.1σ . The dominant uncertainty is the $\sim 40\%$ spread in the local anchor, not the redshift law. We note honestly that an evolving relation is also expected in Λ CDM from halo evolution, so this is *avored over constant*, not a unique confirmation.

z	a_0 (10^{-10} m/s^2)	source
0.00	1.20 ± 0.26	SPARC — McGaugh, Lelli & Schombert 2016
0.05	1.69 ± 0.13	Vărașteanu et al. 2025
0.90	2.38 ± 0.11	MUSE-DARK III 2026

Fit (a₀_decisive_pipeline.py): best $p = 0.80 \pm 0.17$; constant a₀ $\chi^2 = 27$ vs $E(z)$ $\chi^2 = 3.8$ vs $(1+z)^{1.5}$ $\chi^2 = 27.5$. A single clean a₀ at $z = 3$ (3%) sharpens p to $\approx 0.99 \pm 0.04$.

3. The over-constrained web

The content is a single dimensionless invariant, $a_0/cH = 1/Z = 0.173$, held fixed at every epoch and cast by dimensional analysis into six unit-costumes — an acceleration (the radial-acceleration relation), a length ($\lambda = c^2/a_0 = Z \cdot R_H$), a surface density ($\Sigma = a_0/G$), a velocity (the baryonic Tully-Fisher zero-point), a temperature (T_{ds}/Z), and a time (the cosmic free-fall time). From the one premise about a dozen relations follow as forced consequences. The web is *over-constrained*: the same a_0 must simultaneously fit SPARC, the Planck H_0 , the de-Sitter (Λ) floor, the MUSE-DARK measurement at $z \approx 1$, and de Graaff's $z \approx 6$ dynamical masses — five independent determinations pinned by one number, net +4. We use this to draw the honest line: a real relation is z -invariant (a_0/cH stays $1/Z$); a coincidence between cosmic numbers is not. The same test that keeps these edges *rejects* the Standard-Model-constant 'derivations' (Section 5).

4. Three bridges toward a theory of the dark sector

Bridge 1 (relativistic completion — CMB-safety, established). The covariant home is the Aether-Scalar-Tensor theory [Skordis & Złośnik 2021], the one relativistic MOND that fits the CMB. Promoting its fixed a_0 to $a_0(z)$ by coupling to the aether expansion $\theta = \nabla \cdot A = 3H$ is a minimal modification. From the AeST action, a_0 enters only the spatial free-function term $F \propto Y^{3/2}/a_0$; on the FRW background the scalar is temporal, so $Y = O(\delta\varphi^2)$ and the a_0 -term is $O(\delta\varphi^3)$ — absent from the linear equations. Hence running a_0 leaves the linear CMB and matter power spectrum *exactly* invariant. A direct solution of the linear Einstein-Boltzmann system confirms this numerically (the running- a_0 effect is 0) and reproduces the Planck sound horizon $r_s = 144.3$ Mpc and acoustic scale $\lambda_A = 301.7$. The CMB-fitting dust mode lives in the separate temporal sector $K(Q)$ and is a_0 -independent.

Bridge 2 (the coefficient — reported straight, does not close). Horizon thermodynamics derives the evolution $a_0 \propto H$ route-independently and the order of magnitude, but does *not* pin the $O(1)$ coefficient: legitimate matchings give $Z = 1, 2\pi, \approx 6$, and 5.789 , clustering within 8.5% — inside a_0 's $\sim 20\%$ systematic. The framework's 'Schwarzschild' reading is not self-consistent (the enclosed mass is $8\pi/3$ times the Schwarzschild mass for R , so R is not a real horizon). The factor of 2 remains a posit. This costs the framework nothing, because the falsifiable prediction is the *evolution*, which is Z -independent.

Bridge 3 (dark matter = dark energy — sourced, falsifiable). In AeST the single temporal function $K(Q) = -2\Lambda + K_2(Q-Q_0)^2$ carries both the dark-matter-mimicking dust mode ($\rho \propto a^{-3}$) and dark energy (-2Λ) — one function, two terms, tied by the evolving a_0 that floors at the Λ value. The test is a_0 -cosmography: $H(z) = Z \cdot a_0(z)/c$ reconstructs the expansion from galaxy dynamics. It lands on the Planck/local band at $z = 0$ and shows the known $\sim 25\%$ tension at $z \approx 0.9$.

5. Boundaries — what is explicitly not claimed

This is a candidate theory of the *dark sector* (gravity, dark matter, dark energy), not a Theory of Everything. The Standard-Model constants do not connect: a 34,000-formula search hits an arbitrary $O(100)$ target to 0.004% about 20% of the time, 52 of 64 published Z^2 'derivations' contain no Z at all, and a *random* number reproduces α^{-1} and the mass ratios to the same precision as $32\pi/3$. Those retrodictions carry ≈ 0 bits and are not part of this framework. Likewise the coefficient is a posit, and the full CMB peak heights require a dedicated Boltzmann implementation. Stating these limits is what keeps the surviving result trustworthy.

6. The decisive test

One clean deep-MOND measurement of a_0 at $z > 2$ — from resolved JWST/ALMA kinematics or dispersions — pins the exponent p to ± 0.1 , because $E(z)$ is large there ($E(3) \approx 4.6$). It confirms the premise or kills it, independently of the coefficient and of the Hubble tension. That single datum decides the framework.

References

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Reproducibility: `real_research/` — `a0_decisive_pipeline.py`, `REAL_WEB.py`, `bridge1_linear_boltzmann.py`,
`bridge1_aest_equations.md`, `bridge2_coefficient_thermodynamics.py`, `bridge3_dark_sector_unification.py`,
`false_discovery_rate.py`, `can_another_number_do_it.py` (github.com/carlzimmerman/zimmerman-formula).